

RFC: Manifold-Structured Agentic Systems Standardizing Governed Capabilities

Status

Proposed

Authors

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Reviewers

Architecture, Platform, Product, Applied AI

Abstract

This RFC proposes a standard architecture for agentic systems based on the concept of manifold-structured capabilities. It formalizes a shift from advisory agent skills to declarative, governed units of behavior called charts. The goal is to improve reliability, safety, composability, and scalability of AI-driven workflows across the organization.

1. Motivation

Current agent implementations rely heavily on prompts, skills, and best-practice documentation. While effective in isolation, these approaches fail to scale safely due to implicit assumptions, unclear boundaries, and brittle composition. Failures frequently occur at handoffs between skills or when agents operate outside their intended domain. A structural solution is required.

2. Proposal Overview

We propose adopting a manifold-based architecture in which each capability is represented as a chart: a locally valid, declarative specification of behavior. Charts explicitly define when they apply, what they guarantee, and when control must transition elsewhere. A routing layer selects charts based on system state and enforces invariants at runtime.

3. Definitions

Manifold: The full task or problem space the system operates within.

Chart: A locally valid capability with explicit validity, invariants, and boundaries.

Atlas: The collection of charts composing the system.

Transition: A declared handoff between charts with defined state transport.

Invariant: A non-negotiable guarantee enforced during execution.

4. Chart Standard

Each chart **MUST** declare the following sections:

- Validity conditions (when the chart applies)
- Operator (how work is performed locally)
- Invariants (guarantees that must hold)
- Boundary conditions (when to exit)
- Transitions (legal handoffs to other charts)
- Telemetry hooks (for observability)

5. Composition and Overlaps

Charts **MAY** compose only in explicitly declared overlap regions. Overlaps **MUST** define shared representations and invariants. Composition outside declared overlaps is prohibited. This prevents unsafe integration and emergent behavior.

6. Routing and Metrics

A routing layer evaluates system state, selects valid charts, and optimizes execution according to declared metrics such as risk, cost, latency, and safety. Routing decisions **MUST** be deterministic and observable.

7. Adoption Plan

Adoption **SHALL** proceed incrementally:

Phase 1: Convert one high-friction skill into a chart.

Phase 2: Introduce routing and invariant enforcement.

Phase 3: Add overlap protocols and additional charts.

No system rewrite is required.

8. Impact and Benefits

Expected benefits include improved reliability, reduced integration bugs, clearer ownership boundaries, predictable scaling, and enforceable standards. Failures become structural and actionable rather than opaque.

9. Risks and Mitigations

Risk: Over-specification early on.

Mitigation: Start with minimal invariants.

Risk: Chart proliferation.

Mitigation: Treat charts as versioned APIs with owners.

10. Decision Request

Approval is requested to adopt manifold-structured charts as a standard architecture pattern for agentic systems within Atomic 47 Labs, beginning with pilot implementations in selected workflows.